DESCRIPTION OF THE STRATA OF THE QUIVER VARIETY FOR A GENERIC STABILITY PARAMETER IN THE PARABOLIC LEVI EXAMPLE

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Allowed standard prerequisites (editorial).

- The stability parameter zeta^bullet of the parabolic Levi example, the subgraph I^0 and the decomposition of a dimension vector into a part supported on I^0 and the rest. Set up earlier in the same subsection (source lines 1627-1655, the author's subsection on the example of Levi factors of parabolic subgroups). Fixes the objects appearing in the two numbered parts: the stratum Mreg_{zeta^bullet}(v^0,w) x (M_0)_(G-hat), the condition that v - v^0 be supported on I^0, and the weights h_i used in part (2).
- The moduli space M_{zeta^bullet}(v,w) of zeta^bullet-polystable modules and its closed points. Section 2 of the paper (the author's stability and face-structure subsections, source lines 726-1019). Supplies the module (B,a,b)

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representing a closed point and the notion of a zeta^bullet-polystable module used in part (1) of the proof.

Source reference. Original section 2.5: sources/algebra-fl01-quiver-varieties-and-branching/pinned-original-source.tex, source lines 1382–1382. Navigation pointer only: part (2) of the counted proof says the assertion is implicitly proved in that subsection and proceeds to display its own complex here, so the pointer is a locator and not a step of the argument.

Proposition 0.1. (1) *The strata are of the form*

$$\mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v}^0,\mathbf{w}) \times (\mathfrak{M}_0)_{(\widehat{G})}, \tag{0.2}$$

where $\mathbf{v}-\mathbf{v}^0$ *is supported on* I^0*, and* $(\mathfrak{M}_0)_{(\widehat{G})}$ *is a stratum of* $\mathfrak{M}_0(\mathbf{v}-\mathbf{v}^0,0)$ *for the subgraph* I^0*, extended to the whole graph by* 0.

(2) *If* $\mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v}^0,\mathbf{w}) \neq \varnothing$*, we have* $\langle h_i, \mathbf{w}-\mathbf{v}^0\rangle \geq 0$ *for* $i \in I^0$*.*

Proof. (1) A closed point in $\mathfrak{M}_{\zeta^\bullet}(\mathbf{v},\mathbf{w})$ is represented by a $\zeta^\bullet$-polystable module. Let $B \in \mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v}',0)$ be its component with $W=0$. As we have the constraint $\zeta^\bullet \cdot \mathbf{v}' = 0$, our choice of $\zeta^\bullet$ implies that $\mathbf{v}'$ is supported on I^0. Thus $\zeta^\bullet$-stable modules are 0-stable (i.e., simple) modules for the subgraph I^0, extended by 0.

(2) is implicitly proved in Section 2.5. Let us make it more apparent. Consider the complex $\mathscr{C}^\bullet_{kl}$ for a point $[B,a,b] \in \mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v}^0,\mathbf{w})$ and S_i, a module with $\mathbb{C}$ on the vertex $i \in I^0$, and 0 on the other vertices and $B=0$. Explicitly it is given by (see [1, (4.2)]):

$$V_i \xrightarrow{\alpha_i} \bigoplus_{h:\mathrm{i}(h)=i} V_{\mathrm{o}(h)} \oplus W_i \xrightarrow{\beta_i} V_i, \tag{0.3}$$

$$\alpha_i \overset{\text{def}}{=} \bigoplus_{\mathrm{i}(h)=i} B_{\overline{h}} \oplus b_i, \qquad \beta_i \overset{\text{def}}{=} \begin{bmatrix} \sum_{\mathrm{i}(h)=i} \varepsilon(h) B_h & a_i \end{bmatrix}.$$

□

References

[1] Nakajima H., Quiver varieties and Kac–Moody algebras, *Duke Math. J.* **91** (1998), 515–560.

[2] Nakajima H., Quiver varieties and finite-dimensional representations of quantum affine algebras, *J. Amer. Math. Soc.* **14** (2001), 145–238, math.QA/9912158.